

Write Strength, Not Write Count: The β Range Dominates Householder Arity in Gated Delta-Rule Attention

Eric Wu^{1,2}

¹Stanford University

²Alpha AI Engineering

Abstract

Delta-rule linear attention with a channel-wise forget gate can be made more expressive in two ways: by applying more Householder factors per token (*arity*, R), or by widening the range of the write-strength gate β from the strict $(0, 1)$ to the reflection range $(0, 2)$. Recent work has concentrated on the first. We report a small controlled factorial measurement suggesting the second matters far more. In a fully paired 2×2 design crossing $R \in \{1, 2\}$ with β regime, over two non-solvable group word problems (A_5 , S_5) and six seed bundles (48 cells, 3.52 GPU-hours), the β widening at fixed $R = 1$ is large and significant in every position band where both arms are above chance, while doubling R under strict β is consistently positive but roughly two orders of magnitude smaller — and the β widening is *free*, adding zero parameters, whereas doubling R costs +40.3%. The $\beta \times R$ interaction is positive in five of eight de-nested position bands. Because Kimi K3’s KDA layer fixes $\beta = \text{Sigmoid}(\cdot) \in (0, 1)$, a determinant argument shows K3-style nodes cannot obtain a negative-determinant transition at any R ; whether they would benefit from $R > 1$ in practice is a separate question our scale cannot settle. We report four results against our own interest. Our pre-registered mechanism — that reflections help because S_5 ’s odd generator demands $\det = -1$ — is *contradicted*, and the fallback explanation we offer is contradicted too. Our R arms are not parameter-matched. *No* cell survives Holm correction at $n = 6$, and only one does after we correct a metric artifact. And an asymmetric optimization failure (10 of 24 strict- β runs end with unconverged loss, versus 0 of 24 reflection runs) means we cannot separate an expressivity effect from a trainability one. We report the effect, its size, and the four reasons it is not yet established.

1 Introduction

Linear-attention sequence models trade the quadratic cost of softmax attention for a fixed-size recurrent state, and the delta rule [6, 8] has become the dominant way to write into that state: each token performs a rank-one *erase* followed by a rank-one *write*. A well-documented consequence is limited state-tracking ability. Two remedies have been proposed. DELTAPRODUCT [7] increases the number of Householder factors applied per token, a quantity we call the *arity* R . Separately, Grazzi et al. [3] showed that admitting *negative* eigenvalues in the state transition unlocks state tracking that is unreachable otherwise; in a delta-rule node this is achieved by widening the write-strength gate β from the strict range $(0, 1)$ to the reflection range $(0, 2)$, since $1 - \beta$ is the non-unit eigenvalue of each factor.

These two levers are not independent, and the literature has largely pursued the first. That matters practically, because arity is expensive — each extra factor needs its own key and gate projections — while widening β is a one-character change to a sigmoid that adds no parameters at all. It also matters now, because the largest deployed delta-rule attention variant, the Kimi Delta Attention (KDA) layer of Kimi K3 [5], specifies $\beta_t = \text{Sigmoid}(W_{\beta}x_t) \in (0, 1)$ — strictly inside the non-reflective range.

We test this with a fully paired 2×2 factorial crossing ar-

ity $R \in \{1, 2\}$ with β regime at otherwise identical geometry, on two non-solvable group word problems, with six seed bundles per cell. The design is deliberately narrow: it is built to estimate one interaction term rather than to establish a state of the art. It is also small, and we have tried to be scrupulous about what a 48-cell, $n = 6$, 1.4M-parameter probe can and cannot support. Two things in particular govern how our results should be read, and we put them here rather than in a late limitations section.

The metric is a prefix average, and we analyze around it. Our harness scores mean token accuracy over *all* positions $1..L$ of each evaluation sequence. Consequently the five nominal “lengths” are nested, and a model that is correct below $L = 40$ and at chance above it still produces a smooth, gracefully-decaying curve that looks like partial extrapolation. For our strict- β arms this is exactly what happens: a zero-parameter model asserting “correct below 40, at chance above” reproduces their entire length curve to within 1.3pp (Table 3). We therefore report *de-nested position bands* as our primary analysis (Section 5.1) and treat the prefix-averaged numbers as secondary. De-nesting strengthens our A_5 result and *eliminates* two of the seven apparently significant S_5 cells, and we report both directions.

Both strict- β arms are at chance beyond $L = 40$. In every de-nested band past position 40, R1 and DP2-strict score within $2\times$ chance (Table 2). This has a consequence we must state plainly, because it defeats an argument we would otherwise have liked to make: the strict-regime arity contrast cannot bound what capacity is worth, since it is measured between two models that are both floored. The parameter confound described in Section 7 is therefore *open*, not bounded.

Contributions. (i) A controlled factorial measurement of arity against β range at matched geometry, fully paired at the seed-bundle level so every arm within a bundle is scored on a byte-identical evaluation bank (Section 4). (ii) The finding that in every band where both arms are measurable, the β effect exceeds the arity effect by one to two orders of magnitude at zero parameter cost (Section 5.3), together with a positive $\beta \times R$ interaction in five of eight de-nested bands — of which one survives Holm correction (Section 5.1). (iii) A determinant observation about K3-style nodes, and a correction to a tempting but wrong reading of K3’s new bounded-decay parameterization (Section 3). (iv) Two negative results against our own hypotheses: the group-parity account of *why* reflections help is contradicted, and so is the headroom account we offered as a replacement (Section 6). (v) An explicit statement of the trainability confound that we cannot resolve, and of what would resolve it (Section 7).

We claim no new operator and no new theory. The gated Householder operator we measure is already implemented in public kernels and published at scale [2]; the determinant argument of Section 2.3 is a special case of a published result [3], and weaker than it. What is new is the controlled measurement. This is an empirical, partly negative result and we frame it as one.

2 Background

2.1 The gated delta rule

Let $S_t \in \mathbb{R}^{d_k \times d_v}$ be the recurrent state, k_t the (L2-normalized) key, v_t the value, β_t the write strength and $\alpha_t \in (0, 1)^{d_k}$ a channel-wise forget gate. The gated delta rule is

$$S_t = (I - \beta_t k_t k_t^\top) \text{Diag}(\alpha_t) S_{t-1} + \beta_t k_t v_t^\top, \quad (1)$$

with readout $\tilde{o}_t = S_t^\top q_t$. Equation (1) is exactly the KDA recurrence of Kimi K3, and specializes to gated DeltaNet [9] and, with α scalar per head, to Mamba-2-style gating [1].

2.2 Householder arity

DELTAPRODUCT [7] applies R rank-one factors per token, each with its own key and write strength:

$$S_t = \left[\prod_{i=1}^R (I - \beta_t^{(i)} k_t^{(i)} k_t^{(i)\top}) \right] \text{Diag}(\alpha_t) S_{t-1} + \text{writes}. \quad (2)$$

Each factor is a true Householder reflection when $\beta = 2$.

Notation. We write R for the number of factors per token and call it *arity*. DELTAPRODUCT writes n_h , denoting models DELTAPRODUCT $_{n_h}[\cdot]$ with the admissible eigenvalue range in

brackets, and does not use the word “arity”; our R is their n_h , and our arms are their $n_h \in \{1, 2\}$ (their $n_h=1$ is DeltaNet). Arity is not free: the $R=2$ node needs a second set of key and gate projections.

2.3 The determinant fact, and to whom it belongs

The per-token map acting on S_{t-1} in Equation (2) is $A_t = \left[\prod_i (I - \beta_t^{(i)} k_t^{(i)} k_t^{(i)\top}) \right] \text{Diag}(\alpha_t)$. For a unit-norm key, $\det(I - \beta k k^\top) = 1 - \beta$, so

$$\det A_t = \left[\prod_{i=1}^R (1 - \beta_t^{(i)}) \right] \cdot \prod_j \alpha_{t,j}. \quad (3)$$

Since $\alpha_{t,j} > 0$ by construction, the *sign* of $\det A_t$ is governed entirely by the β factors, and if every $\beta^{(i)} \in (0, 1)$ then $\det A_t > 0$ for every R : raising arity cannot recover a negative determinant that the β range has excluded.

What this does *not* predict. We are careful here, because Equation (3) does not by itself predict the interaction we go on to measure. At $R = 2$ with $\beta \in (0, 2)$, two negative factors multiply to a *positive* determinant, so $R=2$ reflection unlocks no determinant sign that $R=1$ reflection cannot already reach. Read strictly, Equation (3) predicts a main effect of β and *no* $\beta \times R$ interaction at $R = 2$. Our measured interaction is therefore not explained by this argument, and we do not present it as confirmed theory. The determinant argument licenses exactly one claim — that strict β forecloses negative determinants at every arity — and we restrict ourselves to that.

Attribution. Grazzi et al. [3] study $\mathcal{M}_k^n(\Omega)$, the products of exactly k generalized Householder matrices $I - \beta_i v_i v_i^\top$ with $(1 - \beta_i) \in \Omega$, $\|v_i\| = 1$. Their Proposition 1.3 states that any eigenvalue of $N \in \mathcal{M}_k^n((-1, 1])$ is either 1 or has modulus below 1, and that *if in addition* $N \in \mathcal{M}_k^n([0, 1])$ and $k \leq 2$, then it lies in $[0, 1] \subset \mathbb{R}$. The $k \leq 2$ hypothesis is essential — it was added in a published correction, and its failure for $k \geq 3$ is what DELTAPRODUCT exploits. They conclude such transitions are the identity or non-orthogonal, hence neither reflections nor rotations, and that for $k \leq 2$ the model cannot learn parity by their Theorem 1. Their Proposition 1.2 is the permutation statement proper: realizing a permutation matrix requires the eigenvalue set $\{-1, 1\}$. Equation (3) is a determinant-level restatement, for the delta rule with normalized keys, of a spectral fact they establish in far greater generality; it also says strictly less, since at $R \geq 3$ a positive determinant no longer implies non-negative real eigenvalues. We claim no theoretical novelty for it whatsoever.

3 The K3 KDA node

Kimi K3 [5] is a mixture-of-experts model with 2.78T total and 104.2B activated parameters, a 1M-token context, and native vision. Each block places three KDA layers followed by one gated MLA layer — a 3:1 ratio *per block* (§2.1) — with one additional MLA layer at the end of the backbone, giving 69 KDA and 24 MLA layers (Table 1, §3.2) and a global ratio of 2.875:1. K3 uses no explicit positional encoding (§3.4) and

applies NoPE to all MLA layers, so positional and recency information reaches the model only through KDA’s gating and decay (§2.1.2). The KDA recurrence is Equation (1), and its write strength is

$$\beta_t^h = \text{Sigmoid}(W_{\beta}^h x_t) \in (0, 1), \quad (4)$$

a plain sigmoid, per head and per token, with no scale factor: we searched the report and β is never scaled, doubled, or rescaled anywhere, with the open interval stated in two independent places (§2.1.1). The remaining projections are $q, k = \text{L2Norm}(\text{Swish}(\text{ShortConv}(W_{q/k} x)))$, $v = \text{Swish}(\text{ShortConv}(W_v x))$, and a decay logit $z = W_{\alpha}^{\uparrow} W_{\alpha}^{\downarrow} x + b_{\alpha}^h$ from a low-rank projection plus a per-head bias.

By Equation (3), K3’s KDA layers have $\det A_t > 0$ at every token and head, and this would remain true at any R . That is a statement about what strict β forecloses, and it holds independently of our experiment. Whether a K3-style node would *in practice* benefit from $R > 1$ is an empirical question that our 1.4M-parameter probe cannot settle; we return to this in Section 7. We note the report contains no Householder, arity, or DELTAPRODUCT discussion, and no analysis of KDA’s state-tracking properties; the question we ask is not one K3 addresses.

A correction worth making explicitly. K3 introduces a *bounded* decay parameterization (§2.1.1, Eq. 5), $g = g_{\min} \cdot \text{Sigmoid}(e^{A_h} z)$ with $g_{\min} = -5$ fixed and A_h a learnable per-head log-scale initialized to 0, so $\alpha = \exp(g) \in (e^{g_{\min}}, 1)$. K3 reports this replaces the unbounded negative-softplus mapping $g = -e^{A_h} \text{Softplus}(z) \in (-\infty, 0)$ used by Kimi Linear [4]. It is tempting to read this floor as what forecloses reflections. It is not. Because $\alpha = \exp(g) > 0$ for any finite g , $\det \text{Diag}(\alpha) > 0$ under *both* parameterizations, so the floor cannot change the sign of $\det A_t$ at all. Determinant sign is governed solely by β . What forecloses odd permutations in K3 is the sigmoid-parameterized β of Equation (4), which Kimi Linear already had.

K3’s stated motivation for the floor is numerical, not representational. The reciprocal cumulative decay $1/\Gamma$ used to rescale keys within a chunk can grow without bound and overflow in finite precision; with $g_{\min} = -5$ the cumulative log-decay over a 16-token tile is bounded, so $1/\Gamma$ stays within BF16 range and *all* causal tiles can use dense Tensor Core matrix multiplications, eliminating the position-pair diagonal path (Fig. 3). No modelling or quality justification is offered, and the report contains no architecture ablations at all — so this is an absence of evidence, not evidence of absence. K3 does note the bounded form is related to lower-bounded recurrence gates in prior work; we read that as a relatedness remark appended after the numerical argument, since the paragraph’s topic sentence and the figure caption are both purely computational.

One consequence K3 does not discuss cuts against our own framing: the floor $\alpha > e^{-5} \approx 6.7 \times 10^{-3}$ is a genuine modelling constraint, since near-total single-step forgetting was representable under Kimi Linear’s unbounded range and is

not under K3’s. Given that KDA is also K3’s only source of positional information, the decay range is load-bearing in a way the numerical framing does not convey. We raise it rather than let it be raised against us.

4 Experimental setup

Arms. We cross arity $R \in \{1, 2\}$ with β regime, giving four arms (Table 1). All use the same gated Householder mixer with the same Triton backend, 4000 steps, batch 64, and no feed-forward blocks. The β regime is a single scale on the sigmoid, $\beta = s \cdot \text{Sigmoid}(z)$ with $s \in \{1, 2\}$; it adds no parameters, and the two regimes have byte-identical parameter ledgers. Post hoc, the strict arms reach $\beta_{\max} = 1.0000$ and the reflection arms $\beta_{\max} = 2.0000$, with no record violating its bound.

Tasks. Two group word problems: given a sequence of generators of a finite group, predict the composed element. We use A_5 (order 60, non-solvable, *both* generators even) and S_5 (order 120, non-solvable, transposition generator *odd*). Chance accuracy is $1/60 = 1.67\%$ and $1/120 = 0.83\%$ respectively. The parity difference motivated the prediction Section 6 reports as falsified.

The metric, stated explicitly. Accuracy is the fraction of correct predictions averaged over *all* L positions of 64 evaluation sequences ($N = 64L$ scored tokens; we verified all 240 reported values are integer multiples of $1/64L$). Training words have length 3–40. Because the metric averages positions 1.. L , the nominal lengths are *nested*: $\text{acc}(512)$ includes positions ≤ 40 that are in-distribution. We therefore define *de-nested bands*

$$\text{band}(L_{i-1}, L_i) = \frac{L_i \text{acc}(L_i) - L_{i-1} \text{acc}(L_{i-1})}{L_i - L_{i-1}}, \quad (5)$$

the mean accuracy on positions in $(L_{i-1}, L_i]$ alone, and use these as the primary analysis. Only bands past position 40 are extrapolation.

Pairing, and why $n = 6$ helps. Six seed bundles, 1101–1106, each fixing four independent seed streams (initialization, data, task, evaluation), pairwise distinct in every record. Within a bundle all four arms are scored on the same evaluation bank: we confirmed one distinct `eval_bank_sha256` per (task, bundle) shared by all four arms, and twelve mutually distinct hashes across the twelve groups. Every contrast is a within-bundle difference cancelling evaluation-sampling noise identically in all arms. There were 0 train/eval collisions.

Statistics. For each cell we compute per-bundle differences and aggregate those. The interaction is differenced *per bundle before averaging*, $i_b = d_b^{\text{refl}} - d_b^{\text{strict}}$, preserving pairing; computing it from arm means with an unpaired standard error would be wrong. We report mean, $\text{se} = \text{sd}/\sqrt{6}$ (sd with $\text{ddof} = 1$), $t = \text{mean}/\text{se}$, and the one-sided 95% lower bound

Table 1: The four arms. “Mixer params” excludes both the token embedding and the output head; it is task-invariant, whereas total parameters differ by task because the head does (+39.7% on A_5 , +39.1% on S_5 from $R=1$ to $R=2$). The β regime is parameter-free. $\bar{\beta}$ is the mean realized β pooled over both tasks and all six bundles.

Arm	β range	R	Mixer params	$\bar{\beta}$
R1	(0, 1)	1	998,092	0.899
DP2-strict	(0, 1)	2	1,400,524	0.779
R1-refl	(0, 2)	1	998,092	1.800
Reflection	(0, 2)	2	1,400,524	1.658

$L95 = \text{mean} - t_{0.95,5} \text{ se}$, $t_{0.95,5} = 2.015$. We also report Holm-corrected outcomes across the band family, since we test many cells.

Compute and provenance. All 48 cells ran on one NVIDIA A10G, 48/48 completing, for 3.52 GPU-hours (cells 228–305 s). All 48 carry one probe-source revision, 0 records were rejected, 0 non-finite loss steps occurred.

5 Results

5.1 Accuracy, and what the prefix metric hides

Table 2 gives both views. On the prefix-averaged metric the dominant feature is the β contrast: at $L = 128$ on A_5 , moving from strict to reflection β at fixed $R = 1$ takes accuracy from 25.74% to 82.83%, while doubling R at fixed strict β takes it from 25.74% to 27.75%.

De-nesting sharpens this and also disciplines it. In every extrapolation band, both strict arms sit within $2\times$ chance — R1 on A_5 scores 1.24–1.91% against 1.67% chance. The reflection arms are genuinely above chance for longer: Reflection on A_5 reaches 96.62% on positions 65–128 and 27.25% on 257–512. Table 3 makes the point quantitatively: a zero-parameter “correct below 40, at chance above” model reproduces both strict arms’ entire length curves to within 1.3pp, and fails on the reflection arms by 12–66pp. The honest reading is that *the strict- β arms do not extrapolate past the training range at all*, and the reflection arms partly do.

This also corrects an interpretation we would otherwise have offered. The strict-regime arity effect on the prefix metric decays monotonically with length (A_5 : +3.31, +3.50, +2.01, +0.98, +0.35pp), which invites a story about extra factors mattering less as the task hardens. It is dilution: the pure prediction $(40/L) \times \text{effect}(40)$ gives S_5 +2.67, +1.67, +0.83, +0.42, +0.21 against observed +2.67, +1.92, +0.82, +0.38, +0.21. That curve is one number divided by L , and we withdraw any interpretation of its shape.

5.2 The interaction

Table 4 gives the interaction under both metrics. On de-nested bands it is positive at uncorrected L95 in five of eight bands, from +22.14 to +57.09pp. De-nesting *strengthens*

A_5 (+13.47 $\rightarrow$ +30.14 at 65–128; +35.28 $\rightarrow$ +57.09 at 129–256) and *eliminates* the two long-length S_5 cells, which reverse sign (+8.57 $\rightarrow$ −0.77; +4.44 $\rightarrow$ +0.31). Those two cells were significant only because prefix averaging carried the 41–64 signal forward; their per-bundle interaction values correlate at $\rho \geq 0.99$ with the $L = 64$ cell. We do not count them.

We must be equally direct about multiplicity. Across the ten prefix cells, *no* cell survives Holm or Bonferroni correction — Holm stops at the first comparison ($p = 0.00658$ against a threshold of 0.005). This is structural, not unlucky: with $df = 5$, clearing $\alpha = 0.005$ requires $|t| > 4.03$, and no prefix cell reaches it. On the eight de-nested bands, exactly one survives (S_5 , 41–64, $t = 4.26$, $p = 0.00401$). So the defensible statement is: one band-level interaction is established at family-wise $\alpha = 0.05$, and four more are positive and large but individually uncorrected. The count-based framing “ k of n cells significant” is especially weak here because the cells are not independent (mean within-task cross-length $r = 0.53$, several at 0.99–1.00).

5.3 β dominates arity, and it is free

Table 5 compares the two levers in the bands where both arms are measurable. On A_5 positions 41–64, widening β at fixed $R = 1$ buys +97.98pp ($t = 87.75$) against arity’s +3.81pp ($t = 2.40$), a ratio of $25.7\times$; on 65–128, +64.44 against +0.52, or $124.7\times$; on S_5 41–64, +30.64 against +0.67, or $45.5\times$. Beyond those bands the comparison stops being meaningful because all four arms approach chance, and we report no ratio there. On the prefix metric the corresponding ratios run $6.3\times$ to $44.9\times$ across all ten cells, but we now regard those as contaminated and give them only for continuity.

The cost asymmetry is the part we consider most robust, because it does not depend on the effect size at all: the β widening adds *zero* parameters — both arms have 998,092 mixer parameters and byte-identical ledgers — whereas doubling arity costs +40.3% of mixer parameters (+39.7% / +39.1% of total, by task). Whatever the true effect sizes, one intervention is free and the other is not.

We do not claim arity does nothing. An earlier framing of this work said arity’s effect was “not distinguishable from zero”. That is true cell-by-cell and misleading overall. The strict-regime arity effect is positive in 10 of 10 prefix cells (one-sided sign test $p = 0.00098$, more significant than any single interaction cell we report), and pooling all 60 per-bundle differences gives +1.61pp, $\text{se} = 0.61$, $t = 2.66$ — though those 60 values are nested and not independent, so the pooled test overstates its own precision. Our design also had little power: the minimum detectable effect at 80% power is 7.4–12.5pp at short lengths, so a real +3.3pp arity effect would have been invisible to us. The accurate statement is that *arity’s benefit under strict β is consistently positive, one to two orders of magnitude smaller than the β effect, and below this design’s per-cell resolution* — not that it is absent.

Table 2: Mean accuracy (%), $n = 6$ bundles. **Left:** the prefix-averaged metric our harness reports, over positions 1.. L . **Right:** the same data de-nested into position bands via Equation (5), which is what actually measures behaviour at new lengths. Chance is 1.67% (A_5) and 0.83% (S_5); *italic* marks band values within $2\times$ chance. Both strict arms are at chance in every extrapolation band — the prefix metric disguises this as graceful decay.

Arm	Task	Prefix-averaged, positions 1.. L					De-nested bands, positions in $(L_{i-1}, L_i]$				
		40	64	128	256	512	1–40	41–64	65–128	129–256	257–512
R1	A_5	79.19	49.96	25.74	13.83	7.82	79.19	<i>1.24</i>	<i>1.53</i>	<i>1.91</i>	<i>1.81</i>
DP2-strict	A_5	82.49	53.45	27.75	14.81	8.17	82.49	5.05	<i>2.04</i>	<i>1.87</i>	<i>1.53</i>
R1-refl	A_5	99.99	99.70	82.83	46.47	23.63	99.99	99.22	65.97	10.11	<i>0.78</i>
Reflection	A_5	100.00	100.00	98.31	82.73	54.99	100.00	100.00	96.62	67.15	27.25
R1	S_5	74.06	46.98	23.65	12.51	6.63	74.06	<i>1.86</i>	<i>0.31</i>	<i>1.37</i>	<i>0.76</i>
DP2-strict	S_5	76.73	48.90	24.47	12.89	6.84	76.73	<i>2.53</i>	<i>0.04</i>	<i>1.31</i>	<i>0.79</i>
R1-refl	S_5	97.07	72.86	37.30	19.58	9.99	97.07	32.50	<i>1.75</i>	<i>1.85</i>	<i>0.41</i>
Reflection	S_5	98.09	88.45	56.04	28.53	14.64	98.09	72.40	23.62	<i>1.02</i>	<i>0.75</i>

Table 3: Why we de-nest. Maximum absolute error, over $L \in \{64, 128, 256, 512\}$, of the *zero-free-parameter* prediction $\text{acc}(L) = \frac{40}{L} \text{acc}(40) + \frac{L-40}{L} \cdot \text{chance}$, i.e. “correct below 40, at chance above”. The strict arms are fit to within 1.3pp: their apparent length curve carries no extrapolation signal. The reflection arms deviate by 12–66pp, so they genuinely do extrapolate.

Arm	A_5 max err	S_5 max err
R1	0.16	0.38
DP2-strict	1.27	0.64
R1-refl	50.44	11.87
Reflection	65.92	26.84

6 Two failed explanations

We stated a mechanistic prediction before running the sweep. It failed. The replacement we then offered also fails against the same data. We report both.

The parity prediction, and its falsification. Equation (3) says reflection β unlocks negative determinants, i.e. odd permutations. Our tasks differ exactly in whether odd permutations are demanded: S_5 ’s transposition generator is odd, so composing a word requires elements of determinant -1 ; in A_5 both generators are even. We predicted a *large* interaction on S_5 and a *small* one on A_5 . We observe the opposite in the bands that matter: de-nested, the interaction is $+30.14$ and $+57.09\text{pp}$ on A_5 at 65–128 and 129–256, against -0.77 and $+0.31\text{pp}$ on S_5 in the same bands. Only the 41–64 band favours S_5 . The reachability-of-reflections story does not explain our result. As Section 2.3 notes, it never should have: Equation (3) predicts no interaction at $R = 2$ in the first place.

The headroom explanation, also contradicted. We then proposed that $\beta > 1$ helps by making each write more expressive regardless of parity, with A_5 (order 60) simply easier than S_5 (order 120) so its reflection arms retain headroom where S_5 ’s have collapsed. This predicts the interaction should track available headroom. It does not. Across the

ten prefix cells, the correlation between interaction and headroom ($100 - \text{acc}$ of R1-refl) is only $r = +0.53$; within A_5 it fits well ($r = +0.93$) but within S_5 there is no relationship ($r = +0.28$, $\rho = +0.10$), and using the Reflection arm’s de-nested band accuracy as the regressor gives the *wrong sign* ($r = -0.33$). The specific claim we had drafted — that the S_5 reflection arms at 19.58% “have little room left” — is also indefensible: 19.58% against a 0.83% chance floor is $\sim 80\text{pp}$ of headroom, more than A_5 at $L = 256$ (46.47%), where the interaction is four times larger.

Where that leaves the mechanism. We have a measured interaction, a falsified pre-registered mechanism, a falsified fallback, and a determinant argument that predicts no interaction at $R = 2$. We therefore offer *no* mechanism. A reader should treat the interaction as an unexplained empirical regularity, and should weigh it against the trainability account in Section 7, which we cannot rule out. Distinguishing the candidates needs a solvable-group control, a difficulty-matched non-solvable pair, and an $R = 3$ arm (where Equation (3) does predict something, since the $k \leq 2$ hypothesis of Grazi Prop. 1.3 fails there). We ran none of these.

7 Limitations

7.1 The trainability confound, which we cannot resolve

This is the objection we consider most serious. Ten of 48 runs end with a final logged loss above 0.1, and *all ten are strict- β* (R1 6/12, DP2-strict 4/12; both reflection arms 0/12). A perfectly asymmetric stability failure between the arms being compared is the signature of an optimization difference, not only an expressivity one. Within the 24 strict runs, log final loss correlates with accuracy at $r = -0.43$ ($L = 40$), and spiked runs are worse at every length.

The confound is load-bearing. Restricting the interaction to bundles where all four arms ended below 0.1 (A_5 : 1102–1105, $n = 4$; S_5 : 1104–1105, $n = 2$), *zero* of ten cells remain significant. The A_5 point estimates survive largely intact ($+13.56$, $+28.52$, $+26.93$ at $L = 128/256/512$ against $+13.47$, $+35.28$, $+31.01$), which is the best we can say for

Table 4: The $\beta \times R$ interaction, differenced per bundle before averaging. **Left:** on the prefix-averaged metric. **Right:** on de-nested position bands, our primary analysis. L95 is the one-sided 95% lower bound ($t_{0.95,5} = 2.015$); **bold** marks L95 > 0 uncorrected. Seven of ten prefix cells and five of eight bands are positive at uncorrected L95. Under Holm correction across the family, **zero** of the ten prefix cells and **one** of the eight bands survive (marked †): with $df = 5$, $p < 0.00625$ needs $|t| > 4.03$ and our maximum band t is 4.26. Note S_5 at 129–256 and 257–512 *reverse sign* once de-nested — they were significant only by inheriting the 41–64 signal through prefix averaging.

Prefix-averaged, positions 1..L					De-nested bands				
Task, L	Inter.	se	t	L95	Task, band	Inter.	se	t	L95
A_5 , 40	−3.30	4.25	−0.78	−11.87	A_5 , 41–64	−3.03	1.53	−1.97	−6.12
A_5 , 64	−3.20	3.09	−1.03	−9.43	A_5 , 65–128	+30.14	9.34	3.23	+11.32
A_5 , 128	+13.47	4.74	2.84	+3.93	A_5 , 129–256	+57.09	16.08	3.55	+24.69
A_5 , 256	+35.28	9.38	3.76	+16.37	A_5 , 257–512	+26.74	12.56	2.13	+1.43
A_5 , 512	+31.01	10.70	2.90	+9.45	S_5 , 41–64	+39.23†	9.21	4.26	+20.66
S_5 , 40	−1.65	2.42	−0.68	−6.52	S_5 , 65–128	+22.14	8.80	2.52	+4.40
S_5 , 64	+13.68	4.23	3.23	+5.15	S_5 , 129–256	−0.77	0.79	−0.98	−2.35
S_5 , 128	+17.91	6.33	2.83	+5.15	S_5 , 257–512	+0.31	0.24	1.31	−0.17
S_5 , 256	+8.57	3.19	2.69	+2.15					
S_5 , 512	+4.44	1.68	2.64	+1.05					

Table 5: β range versus arity, on de-nested bands, restricted to the three extrapolation bands where the compared arms are above chance. Both interventions start from the same $R=1$ strict baseline; Δ is a paired mean difference in points. The β intervention is parameter-free; arity costs +40.3% mixer parameters. Bands past these are omitted because all arms approach chance, making the ratio meaningless.

Task, band	Lever	Δ	t	Ratio
A_5 , 41–64	β	+97.98	87.75	
A_5 , 41–64	arity	+3.81	2.40	25.7×
A_5 , 65–128	β	+64.44	7.55	
A_5 , 65–128	arity	+0.52	1.36	124.7×
S_5 , 41–64	β	+30.64	6.05	
S_5 , 41–64	arity	+0.67	1.02	45.5×

the effect; the significance does not. Traces make the problem concrete: R1 on S_5 bundle 1106 ends at 0.85 having reached 10^{-4} at step 3000, and bundle 1103 oscillates between ~ 0 and 1.27 across the last 1500 steps. Those are not converged models. Since $\beta = 2$ doubles the effective write-gate gradient scale, the reflection arms’ perfect stability record may itself be an optimization-geometry effect.

We therefore cannot separate “reflection β is more expressive” from “reflection β is easier to optimize in 4000 steps”. Our loss traces are logged only every 500 steps, too coarse to characterize the spikes, and we retained no intermediate checkpoints, so we cannot re-evaluate from a pre-spike model. Resolving this needs a learning-rate sweep per arm, longer training, or best-checkpoint evaluation. Until then, this paper measures *the joint effect of expressivity and trainability* of the β range, and the reader should read every result above with that substitution.

7.2 The R arms are not parameter-matched, and the confound is open

The $R=1$ arms have 998,092 mixer parameters and the $R=2$ arms 1,400,524, +40.3%. Any raw “ $R=2$ beats $R=1$ ” main effect is confounded with capacity and we claim none.

We had intended to argue that the strict-regime arity effect ($\leq +3.50\text{pp}$) upper-bounds what that capacity is worth. We withdraw that argument. As Table 2 shows, both strict arms are at chance in every extrapolation band, and a capacity contrast measured between two floored models is guaranteed to read ≈ 0 regardless of what capacity is worth. The bound is uninformative rather than reassuring, and using it would have been circular. Two further problems: the DP2-strict vs R1 contrast varies arity and capacity together, so if arity were mildly harmful under strict β while capacity helped, the observed difference would bound neither component; and the design’s minimum detectable effect is 7.4–12.5pp, so capacity could be worth up to $\sim +11.9\text{pp}$ within our confidence bounds.

It remains true that capacity differs by the same 402,432 parameters in both regimes, so to first order it cancels from a difference of differences. But that argument assumes the *value* of capacity is regime-independent, and we never varied capacity within the reflection regime, so we cannot check it. The fix is specified: the project defines an R1-P arm feed-forward-width-matched to DP2-strict at 1,399,756 parameters (a 0.055% match), which was *not* run. Running it plus a reflection-regime counterpart — roughly 24 cells at a cost comparable to this sweep — would isolate capacity within each regime. That control is necessary, not optional, and its absence is what prevents us from making an arity claim.

7.3 $n = 6$, multiplicity, and influential bundles

Six bundles makes paired contrasts informative but imprecise: interaction standard errors run 1.68–16.08pp, and the Reflection arm’s between-bundle spread on A_5 is large

(sd 20.19pp at $L = 256$). As Section 5.1 states, no prefix cell and only one band survives Holm correction, and with $df = 5$ none could have cleared Bonferroni even in principle. Leave-one-bundle-out shows the two long-length S_5 prefix cells lose significance if either of two bundles is dropped — consistent with their sign reversal under de-nesting. The A_5 result is more robust: dropping the most extreme bundle (1103) takes $L = 256$ from +35.28 to +29.18 (L95 +10.56) and $L = 512$ from +31.01 to +22.26 (L95 +6.18), both still positive. We report L95 values, not point estimates, as the defensible claim.

7.4 Ceiling and floor censoring

Of 240 reported values, 26 are exactly 100.00% and 35 exceed 99%; of the de-nested band values, 107 of 240 lie within $2\times$ chance and 10 band-cells fall strictly below it. At $L = 40$ the reflection arms score 99.99% and 100.00% on A_5 , the latter with zero variance across all six bundles, so the $L = 40$ interaction is mechanically bounded and its negative sign (−3.30pp) is not interpretable. As a robustness check we recomputed the interaction on Haldane–Anscombe-corrected log-odds: sign and significance broadly survive (9 of 10 prefix cells positive at uncorrected L95), but the largest interaction moves from $L = 256$ to $L = 128$ on A_5 , and S_5 at $L = 40$ becomes significantly *positive*. So the raw-scale magnitudes are partly an artifact of pp differences being maximized near 50%, and our reading of the $L = 40$ cells as uninformative is right on A_5 but wrong on S_5 .

7.5 Scale, scope, and the K3 claim

Our models have 1.0–1.4M mixer parameters, no feed-forward blocks, and train for 4000 steps on synthetic group word problems. Kimi K3 has 2.78T total parameters. **Nothing in this paper is direct evidence about the behaviour of a frontier-scale model.** We separate the two K3 statements accordingly. The determinant observation — that strict β forecloses negative-determinant transitions at every arity — follows from Equation (3) alone, is regime-agnostic, and does not depend on this experiment; we consider it sound. The empirical prediction that K3-style nodes would not *benefit* from $R > 1$ does depend on this experiment, and given the four caveats above we present it as a hypothesis motivating a scaled replication, not as a finding. Group word problems are also a deliberately adversarial probe chosen because they discriminate; language modelling may weight these mechanisms differently.

We also flag a tension with the prior work we rely on. DELTAPRODUCT reports that its models “fail to learn even the training context-length when restricted to the standard eigenvalue range $[0, 1]$, regardless of the number of Householder transformations n_h ”. Our strict- β arms do learn the training range (74–82% at $L = 40$). Either the tasks differ materially or our strict arms are not the regime they describe; we cannot say which.

7.6 Other limitations

We measure $R \in \{1, 2\}$ only, so we cannot speak to $R \geq 3$ — where, as noted, the theory actually predicts something different. All arms share one mixer implementation and Triton backend, so a kernel bug interacting with the β scale would not be caught by our design. We report a single learning-rate schedule per arm with no per-arm tuning, which is precisely what the trainability confound requires.

8 Related work

Delta-rule linear attention. The delta rule as a fast-weight update goes back to Schlag et al. [6]; DeltaNet [8] made it trainable at scale with a chunkwise-parallel form, and gated DeltaNet [9] added the channel-wise forget gate that makes Equation (1) the modern default. Mamba-2 [1] supplies the structured-state-space view of the same gating. Kimi Linear [4] introduced KDA, and Kimi K3 [5] deploys it as the majority attention layer at frontier scale.

Arity and negative eigenvalues are one research programme. The two lines of work our paper sits between share most of their authors. Grazi et al. [3] established that admitting negative eigenvalues unlocks state tracking otherwise out of reach — the result our Section 2.3 specializes and which subsumes our determinant observation. DELTAPRODUCT [7] then raised arity, and its §5.2 reports the finding that most directly prefigures ours: “Unexpectedly, S_4 and A_5 can extrapolate robustly using only $n_h = 2$ despite the theorem suggesting 3 and 4, respectively.” Two details matter for us. They attribute the efficiency to S_4 and A_5 being isomorphic to subgroups of $SO(3, \mathbb{R})$, and they observe a head that learns $\beta = 2$ on *both* factors — two true reflections composing to a rotation. And their §5.2 uses the extended eigenvalue range throughout, reporting the failure under $[0, 1]$ quoted in Section 7. That sentence is the closest prior statement to our result. Our contribution is to convert it from a training-time observation reported in passing into a paired factorial estimate at length extrapolation — with the caveats above. Our result is a refinement of their programme, not a challenge to it.

The operator is not new, and we are precise about what is prior. The per-channel-gated delta-rule operator we measure is computed by public kernels for generalized (diagonal-plus-low-rank) delta-rule updates, and is established at scale: EDA [2] trains dense 2.5B and MoE 25B-A2.8B models on it — though it appears there as the *baseline*, explicitly credited to KDA, EDA’s own contribution being a second, independently addressed erase factor. To avoid overclaiming on DELTAPRODUCT’s behalf: its Appendix B.4 Eq. 6 does place a per-channel $\text{diag}(w)$ inside a product at arbitrary n_h , but the factors are asymmetric RWKV-7 matrices rather than Householders, the construction is offered explicitly as buying expressivity “at the cost of stability”, and it is used for a theory result with no experiments; DELTAPRODUCT’s own gated variant uses a *scalar* gate outside the product. We contribute

no operator, no kernel and no theorem — only a controlled measurement.

9 Conclusion

In a gated delta-rule node, the write-strength range β and the Householder arity R are not interchangeable routes to expressivity. Across 48 paired cells on two non-solvable group word problems, widening β produced large gains in every position band where the compared arms were above chance — $25.7\times$ to $124.7\times$ the gain from doubling arity — at zero parameter cost, against arity’s $+40.3\%$. The $\beta \times R$ interaction was positive in five of eight de-nested bands. The determinant argument of Section 2.3, which is not ours, establishes independently of any experiment that a strict- β node such as Kimi K3’s KDA layer cannot obtain a negative-determinant transition at any arity.

We are not able to claim more than that, and we would rather say so than imply otherwise. Four things stand between this measurement and a result. Our pre-registered mechanism is falsified and the fallback we offered is falsified too, so we have no mechanism. Our arity arms are not parameter-matched, and the bound we had hoped to place on that confound turns out to be uninformative because both strict arms are at chance. Only one of eight band-level interactions survives multiple-comparison correction at $n = 6$, and none of the ten cells on the prefix-averaged metric does. And an asymmetric optimization failure means what we measured may be trainability rather than expressivity. Each has a specified, cheap fix — the R1-P capacity control, more bundles, an $R = 3$ arm, and per-arm learning-rate tuning with checkpoint retention. What we can defend is narrower than where we started: one lever is free and appears to matter a great deal, the other is expensive and appears to matter little, and we do not yet know why or whether it survives at scale.

References

- [1] T. Dao and A. Gu. Transformers are SSMS: Generalized models and efficient algorithms through structured state space duality. *ICML*, 2024. arXiv:2405.21060.
- [2] X. Li, C. Zhang, H. Luo, X. Lin, Z. Wang, Z. Qiu, Y. Mao, L. Chen, M. Yuan, M. Sun, H. Jiang, S. Zhang, R. Men, W. Hu, G. Cheng, B. Zheng, D. Liu, and J. Zhou. Erase-then-Delta attention: Decoupling erase and write addresses in delta-rule linear attention. arXiv:2606.26560, 2026.
- [3] R. Grazi, J. Siems, A. Zela, J. K. H. Franke, F. Hutter, and M. Pontil. Unlocking state-tracking in linear RNNs through negative eigenvalues. *ICLR*, 2025. arXiv:2411.12537.
- [4] Kimi Team. Kimi Linear: An expressive, efficient attention architecture. arXiv:2510.26692, 2025.
- [5] Kimi Team. Kimi K3: Open frontier intelligence. arXiv:2607.24653, 2026.
- [6] I. Schlag, K. Irie, and J. Schmidhuber. Linear transformers are secretly fast weight programmers. *ICML*, 2021. arXiv:2102.11174.
- [7] J. Siems, T. Carstensen, A. Zela, F. Hutter, M. Pontil, and R. Grazi. DeltaProduct: Improving state-tracking in

linear RNNs via Householder products. *NeurIPS*, 2025. arXiv:2502.10297.

- [8] S. Yang, B. Wang, Y. Zhang, Y. Shen, and Y. Kim. Parallelizing linear transformers with the delta rule over sequence length. *NeurIPS*, 2024. arXiv:2406.06484.
- [9] S. Yang, J. Kautz, and A. Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. *ICLR*, 2025. arXiv:2412.06464.